

DATA STRUCTURE

LAB PROGRAMS

12. a) Queue Insertion (Enqueue) Using Array

```
#include <stdio.h>

#define MAX 10

int main()
{
    int queue[MAX];
    int front = 0, rear = -1;
    int n, i, value;
    printf("Enter number of elements to insert: ");
    scanf("%d", &n);
    for(i = 0; i < n; i++)
    {
        if(rear == MAX - 1)
        {
            printf("Queue Overflow\n");
            break;
        }
        printf("Enter element: ");
        scanf("%d", &value);
        rear++;
        queue[rear] = value;
    }
    printf("Queue elements are: ");
    for(i = front; i <= rear; i++)
```

```

{
    printf("%d ", queue[i]);
}
return 0;
}

```

```

Output

Enter number of elements to insert: 4
Enter element: 36
Enter element: 58
Enter element: 69
Enter element: 79
Queue elements are: 36 58 69 79

=== Code Execution Successful ===

```

b) Queue Deletion (Dequeue) Using Array

```

#include <stdio.h>

#define MAX 10

int main()
{
    int queue[MAX];
    int front = 0, rear, n, i;

    printf("Enter number of elements in queue: ");
    scanf("%d", &n);

```

```
    rear = n - 1;
    printf("Enter queue elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &queue[i]);
    }
    if(front > rear)
    {
        printf("Queue Underflow");
    }
    else
    {
        printf("Deleted element = %d\n", queue[front]);
        front++;
        printf("Queue after deletion: ");
        for(i = front; i <= rear; i++)
        {
            printf("%d ", queue[i]);
        }
    }
    return 0;
}
```

Output

```
Enter number of elements in queue: 5
Enter queue elements:
12 32 54 65 67
Deleted element = 12
Queue after deletion: 32 54 65 67

=== Code Execution Successful ===
```

c) Queue Display Using Array

```
#include <stdio.h>
#define MAX 10
int main()
{
    int queue[MAX];
    int front = 0, rear, n, i;
    printf("Enter number of elements in queue: ");
    scanf("%d", &n);
    rear = n - 1;
    printf("Enter queue elements:\n");
    for(i = 0; i < n; i++)
    {
        scanf("%d", &queue[i]);
    }
    if(front > rear)
```

```
{  
    printf("Queue is Empty");  
}  
else  
{  
    printf("Queue elements are: ");  
    for(i = front; i <= rear; i++)  
    {  
        printf("%d ", queue[i]);  
    }  
}  
return 0;  
}
```

Output

```
Enter number of elements in queue: 3  
Enter queue elements:  
34 46 67  
Queue elements are: 34 46 67  
  
=== Code Execution Successful ===
```